

Alvin Ndungu

Software Engineer · Backend, Full-Stack & ML Research

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PROFESSIONAL SUMMARY

I am a results-driven Software Engineer with 3+ years of experience across the full stack; from scalable backend systems and RESTful APIs to polished mobile, web, and desktop interfaces. I have a proven track record integrating generative AI into production environments and contributing to ML research, including model training, biosimulation tooling, and data sourcing within a healthcare AI context. I am skilled at owning the full product lifecycle from architecture through deployment.

TECHNICAL SKILLS

Languages & Frameworks: Python (Django), Node.js, TypeScript, JavaScript, React, React Native, Rust (Tauri)

ML & Research: Model training & fine-tuning, data sourcing & curation, biosimulation tool setup, scikit-learn, TensorFlow/Keras

Cloud & DevOps: AWS (EC2, S3, Cognito), GCP (Compute Engine, Cloud Storage), Nginx, Gunicorn, pm2, Linux Server Administration

Databases & Architecture: MongoDB, MongoEngine, Firebase Realtime DB, Cloud Firestore, Systems Design

Desktop: Tauri (cross-platform desktop apps via Rust + web frontend)

Tools & Methodologies: REST APIs, Git/GitHub, Agile/Scrum, Product Management, Stripe & RevenueCat Integration

App & Project Management: Google Play Console, Apple App Store Connect, KPI tracking, GitHub Scrum boards, Agile/Scrum, Microsoft Teams Planner

Languages: English (fluent), Swahili (native)

PROFESSIONAL EXPERIENCE

Software Engineer & Product Manager · iZola Limited · August 2023 – Present

Mobile & web application development for healthcare platform Xaidi

- Led end-to-end development of Xaidi, a React Native + Django/Node.js healthcare platform, scaling it to over 18,000 downloads on Google Play and the App Store.
- Architected and maintained RESTful backend services, integrating Stripe and RevenueCat for complex subscription and payroll logic.
- Deployed and managed production infrastructure on AWS EC2 and GCP Compute Engine using Nginx, Gunicorn, and pm2 to achieve consistent uptime.
- Integrated generative AI features including an AI chat assistant directly into the platform, improving user engagement metrics.
- Defined and tracked critical KPIs across the product lifecycle, using data-driven insights to guide feature prioritisation and release planning.
- Developed and maintained web application interfaces using React and TypeScript, complementing the mobile product with a browser-based experience.
- Conducted peer code reviews, enforced best practices, and maintained high code quality standards across the engineering team.

ML Research Engineer · Vitafluence AI · September 2024 – April 2025

Secondment to the core research team; contributed to AI/ML research in healthcare and biosimulation

- Sourced, cleaned, and curated datasets for training and evaluating machine learning models in a healthcare AI context.
- Trained and iterated on ML models, managing experiment tracking and performance evaluation.
- Set up and maintained biosimulation tools as part of the research infrastructure, supporting computational experiments.
- Conducted applied research into AI regulatory compliance and healthcare AI systems, directly informed by EU AI Act requirements.

- Collaborated with the research team to document findings and translate research outputs into actionable engineering recommendations.

PROJECTS & SIDE WORK

Cross-Platform Desktop Development (Tauri) - Built cross-platform desktop applications using Tauri (Rust backend + web frontend), gaining hands-on experience with system-level programming and packaging for macOS, Windows, and Linux.

NSE Finance (Flutter) - Built a cross-platform finance application tracking Nairobi Securities Exchange data, primarily targeting mobile but extended to desktop and web.

CERTIFICATIONS

EU AI Act Essentials · KI-Campus (Stifterverband) · *Issued April 2025*

Covers EU AI Act fundamentals, risk classification, stakeholder obligations (Provider, Deployer, National Authority), implementation timelines, and practical compliance scenarios.

ICDL Certificate · Daystar University · *Issued August 2019*

Covered Microsoft Office suite (Word, Excel, PowerPoint, Outlook), desktop publishing principles, and introductory .NET development for desktop applications.

EDUCATION

B.Sc. Computer Science · Kabarak University - *September 2019 – December 2023*

KCSE - Rongai Agricultural & Technical High School - *2015 – 2018*

KCPE - *December 2014*